Video Transcript

Video 9: CDC – Creating a Database Using Python and MySQL (04:13)

At this point, you have all the knowledge that is needed to be able to create a full change data capture application. The full beast if you would know how to create shell commands, create containers, delete containers, initialize databases, clear out databases, and create time loops. As you can see there, we can do all of this within Python. So, at this point you could go ahead and get started on your own, but it can be a bit overwhelming to tackle the full breadth of this application. So, I'm going to go ahead and get you started, and I will start off here with MySQL and Mongo and then leave it up to you to add Redis and Cassandra. It's always a good idea to start small and then build up. We're going to be using the following architecture. As you can see here, I have a number of files and I've given you also the names that I'm using for these files. I'll start off here from the left and as you can see there, there is going to be a file that is dedicated to containers. Through it, we will be able to create the containers for the databases, delete them, and initialize the databases that require it within them. We're also going to have one called scheduler, and you can see that there in the center of the page. And that file is going to be importing the files that are specific to each of the databases. You can see there that I have one for my SQL, one for Mongo, and so on. The Gray dot ones are the ones that are for you.

Those are the ones that you will write. Now, once we pull those in, as you can see there from the icons above, we will be able to create, read, and update information. We will also add some more logic to be able to clear out the data from the databases. So, simply to emphasize, as you can see here, the containers file, containers will be able to create the lead, initialize the scheduler file will be importing those database specific files, then do a time loop and leverage those functions. So, with that in mind, let's go ahead and move to the code. As you can see here, I have files corresponding to the ones that we just saw on the diagram, and I'm going to start off here with MySQL. This is a database server that we have used quite a bit. So, you're quite familiar with how to connect to a database, how to input information, how to read information. So, as you can see here, I start off by adding, importing the driver and then I have a couple of supporting or a few supporting packages there. Now, the next thing that I do is that I make my connection to the database, create a cursor and then I write within this function.

Note here, that I am creating a unique identifier and then a timestamp which gets pushed into the database of the database called Pluto. You can see that here on line 9 and into the table called posts. Now, it's worth noting that this is what begins the entire cycle of data capture. This is what will propagate through the rest of the databases. But this is the place where it all starts. Now, below that I have a number of additional functions. You can see there that I have the read function. This one will read the last five entries that have been made into that database, and you can see there that that gets packaged into an array, which is then returned to the calling function. I also have a method, a function below that that is simply going to clear out the information in the database. This is a good idea to do when you're working in a pretty high iteration; where you want to be able to make big changes, remove them, and then start over. Now, the last part of that is simply handling the case where we exit and that it closes the connections.